WORKED SOLUTION

STEP I 2010, Question 2

Pure • Curve sketching

STEP I 2010, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q2

The key idea

Differentiate with the quotient rule and pull out the factor e^x, which is never zero. The stationary points are then the roots of a quadratic, so "two stationary points" is a statement about its discriminant. For the sketches, the sign of that same quadratic tells us where the curve rises and falls, and the behaviour near the asymptote and as $x \to \pm\infty$ fills in the rest.

Solution

Write $y = \dfrac{x-a}{x-b}e^x$, defined for $x \neq b$. By the product rule, with $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{x-a}{x-b}\right) = \dfrac{(x-b)-(x-a)}{(x-b)^2} = \dfrac{a-b}{(x-b)^2}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x-a}{x-b}e^x + \frac{a-b}{(x-b)^2}e^x = \frac{e^x[(x-a)(x-b)+(a-b)]}{(x-b)^2}.$$

Since $e^x > 0$ and $(x-b)^2 > 0$ for $x \neq b$, the sign of $\mathrm{d}y/\mathrm{d}x$ is the sign of

$$q(x) = (x-a)(x-b)+(a-b) = x^2 - (a+b)x + ab + a - b,$$

and the stationary points are exactly the roots of q other than $x = b$.

Ask yourself Could a root of q sit at $x = b$, where the curve isn't defined?

No harm can come from this: $q(b) = a - b$, so $x = b$ is a root only when $a = b$. But then the curve is just $y = e^x$ (with a hole at $x = b$), which has no stationary points at all. So in every case the stationary points are precisely the real roots of q, and there are two of them exactly when q has two distinct real roots, that is, when its discriminant is positive:

$$(a+b)^2 - 4(ab+a-b) = (a-b)^2 - 4(a-b) = (a-b)(a-b-4) > 0.$$

A product of two factors is positive when both are positive or both negative. Both negative means $a-b<0$; both positive means $a-b>4$ (which already forces $a-b>0$). Hence two stationary points require

$$a-b<0 \quad \text{or} \quad a-b>4,$$

as required. (Conversely, either condition gives two stationary points.)

Part (i)

With $a=0$, $b=\frac{1}{2}$ we have $a-b=-\frac{1}{2}<0$, and

$$y=\frac{x\,\mathrm{e}^x}{x-\frac{1}{2}}, \qquad q(x)=x^2-\tfrac{1}{2}x-\tfrac{1}{2}=(x-1)(x+\tfrac{1}{2}).$$

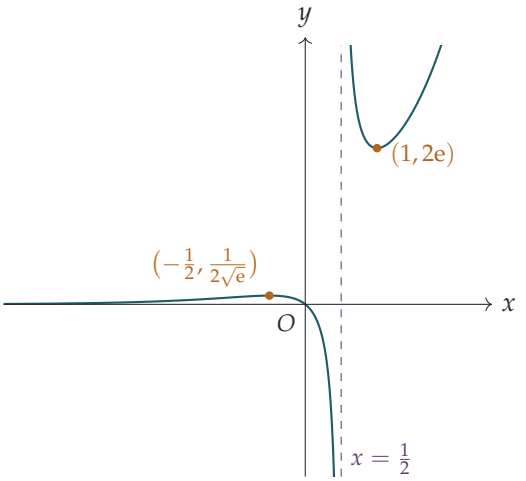
The stationary points are at $x=-\frac{1}{2}$ and $x=1$, one on each side of the vertical asymptote $x=\frac{1}{2}$, as required. Their y-values are

$$y(-\tfrac{1}{2})=\frac{-\frac{1}{2}\,\mathrm{e}^{-1/2}}{-1}=\frac{1}{2\sqrt{\mathrm{e}}}\approx 0.30, \qquad y(1)=\frac{\mathrm{e}}{\frac{1}{2}}=2\mathrm{e}\approx 5.44.$$

Since $q>0$ for $x<-\frac{1}{2}$, $q<0$ for $-\frac{1}{2}<x<1$ (with $x\neq\frac{1}{2}$) and $q>0$ for $x>1$, the point $(-\frac{1}{2},\frac{1}{2\sqrt{\mathrm{e}}})$ is a local maximum and $(1,2\mathrm{e})$ is a local minimum.

Other features:

- As $x\to-\infty$, $\frac{x}{x-1/2}\to 1$ and $\mathrm{e}^x\to 0$, so $y\to 0$. For $x<0$ both x and $x-\frac{1}{2}$ are negative, so $y>0$: the curve approaches the negative x-axis from above.
- The curve meets the axes only at the origin, and $y<0$ for $0<x<\frac{1}{2}$.
- As $x\to\frac{1}{2}^-$, $y\to-\infty$; as $x\to\frac{1}{2}^+$, $y\to+\infty$ (the numerator tends to $\frac{1}{2}\mathrm{e}^{1/2}>0$).
- As $x\to\infty$, $y\to\infty$, growing like e^x.



Part (ii)

With $a = \frac{9}{2}$, $b = 0$ we have $a - b = \frac{9}{2} > 4$, so again there are two stationary points. Now

$$y = \frac{x - \frac{9}{2}}{x}\,\mathrm{e}^x, \qquad q(x) = x^2 - \tfrac{9}{2}x + \tfrac{9}{2} = \tfrac{1}{2}(2x - 3)(x - 3),$$

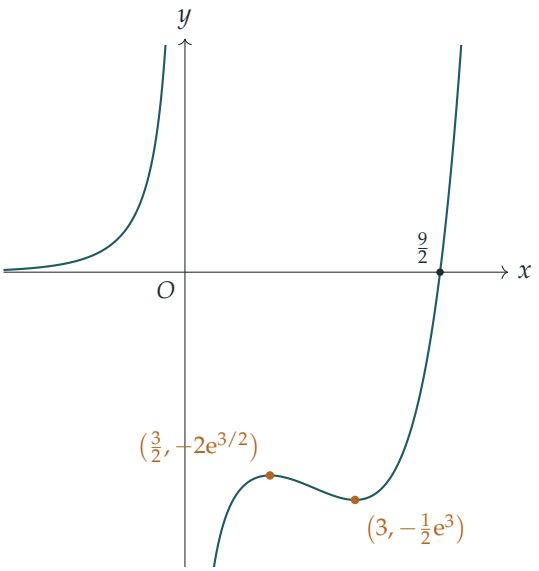
so the stationary points are at $x = \frac{3}{2}$ and $x = 3$, both to the right of the asymptote $x = 0$. Their y-values are

$$y\left(\tfrac{3}{2}\right) = \frac{-3}{\frac{3}{2}}\,\mathrm{e}^{3/2} = -2\mathrm{e}^{3/2} \approx -8.96, \qquad y(3) = \frac{-\frac{3}{2}}{3}\,\mathrm{e}^3 = -\tfrac{1}{2}\mathrm{e}^3 \approx -10.04.$$

The sign of q is positive for $x < \frac{3}{2}$ ($x \neq 0$), negative on $\left(\frac{3}{2}, 3\right)$ and positive for $x > 3$, so $\left(\frac{3}{2}, -2\mathrm{e}^{3/2}\right)$ is a local maximum and $\left(3, -\frac{1}{2}\mathrm{e}^3\right)$ a local minimum. Note that both lie below the x-axis, and the minimum is only slightly lower than the maximum, so the dip between them is shallow.

Other features:

- As $x \to -\infty$, $y \to 0$; for $x < 0$ both $x - \frac{9}{2}$ and x are negative, so $y > 0$. On $x < 0$ the curve is increasing (as $q > 0$ there), from just above the axis up towards the asymptote.
- As $x \to 0^-$, $y \to +\infty$; as $x \to 0^+$, $y \to -\infty$ (the numerator tends to $-\frac{9}{2}$).
- The only intercept is $\left(\frac{9}{2}, 0\right)$; $y < 0$ on $\left(0, \frac{9}{2}\right)$ and $y > 0$ for $x > \frac{9}{2}$, with $y \to \infty$ as $x \to \infty$.



Here the y-axis itself is the vertical asymptote.

Checks

- A numerical derivative of y at random (a, b, x) matches the formula for $\mathrm{d}y/\mathrm{d}x$, and counting sign changes of q for random (a, b) gives two stationary points exactly when $a - b < 0$ or $a - b > 4$.
- In (i), $y(1) = 5.4366 = 2\mathrm{e}$ and $y(-\frac{1}{2}) = 0.3033$; in (ii), $y(\frac{3}{2}) = -8.9634$ and $y(3) = -10.0428$, with values just either side of each point confirming maximum or minimum.
- Sanity check on the condition: in (i) $a - b = -\frac{1}{2}$ and in (ii) $a - b = \frac{9}{2}$, both in the allowed range, as they must be.

Common slips

- Expanding the discriminant without spotting that it is $(a-b)^2 - 4(a-b)$; it factorises neatly once written in terms of $a - b$.
- Forgetting to check that a root of q can't be at the asymptote $x = b$.
- In (ii), drawing the maximum and minimum on opposite sides of the x-axis: both are below it, and the curve crosses the axis only at $x = \frac{9}{2}$.

Beyond STEP

The method here, reading off the shape of a curve from the sign of its derivative and its limits, is made rigorous in a first analysis course through the Mean Value Theorem (a positive derivative on an interval really does mean increasing) and careful definitions of limits at infinity. That e^x beats any rational function as $x \to \pm\infty$ is a standard result proved there too. Abbott's *Understanding Analysis* covers all of this beautifully.

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